🌱 Sustainable Tomorrow

☀️ Light ≡

🔍

Explore raw climate datasets in Excel-style tables. Filter, sort, and export data from NASA, NOAA, IRENA, and NSIDC sources.

📊 Dataset Selection

🌡️ Temperature Data
NASA GISTEMP global temperature anomalies (1880-2024)

⚡ Renewable Energy
IRENA renewable energy capacity data (2000-2024)

⛏️ Critical Minerals
IEA critical minerals demand projections (2024-2050)

🧊 Sea Ice Data
NSIDC Arctic and Antarctic sea ice extent (1979-2024)

🏭 CO2 Data
NOAA CO2 concentrations and ice core data (1880-2024)

📈 Data Statistics

145	23	5
Years measured	Columns	Datasets

🔍 Filter data... (search any column)

📥 Export CSV

🤖 Sustainability Chatbot

🌡️ Temperature Data

Scroll horizontally ↔️ to see all data

Year ↓	Temperature An...	Global Average ...	Arctic Temperat...	Tropical Tempe...	Ocean Tempera...
2,024	1.47	15.47	2.94	1.03	1.18
2,023	1.36	15.36	2.72	0.95	1.09
2,022	1.08	15.08	2.16	0.76	0.86
2,021	1.04	15.04	2.08	0.73	0.83
2,020	1.21	15.21	2.42	0.85	0.97
2,019	1.17	15.17	2.34	0.82	0.94
2,018	1.04	15.04	2.08	0.73	0.83
2,017	1.11	15.11	2.22	0.78	0.89
2,016	1.21	15.21	2.42	0.85	0.97

About This Project

Created for the **Viz Con 2025** contest at **Analyticon**(Amazon-wide Conference on Data Engineering, Science and Analytics). This interactive application explores the connections between human activity and environmental conservation through compelling data narratives using real datasets.

Data Sources

🌡️ **Temperature Data:**
NASA GISTEMP v4 (Goddard Institute)
⚡ **Renewable Energy:**
IRENA Global Energy Statistics
⛏️ **Critical Minerals:**
IEA Critical Minerals Data Explorer
🧊 **Sea Ice Data:**
NSIDC Sea Ice Index
🏭 **CO2 Data:**
NOAA Mauna Loa Observatory

Technology & AI

Frontend: React.js, D3.js, Chart.js, Framer Motion
Backend: Python Flask, NumPy, Pandas
AI Integration: Pattern recognition, correlation analysis, predictive modeling, and natural language insight generation
AI Chatbot: Amazon Bedrock (Nova Pro Model), API Gateway, AWS Lambda, S3, IAM
Deployment: AWS Amplify with automated CI/CD pipeline

Viz Con 2025 Entry

Entry for the **2025 Viz Con contest**, part of the **Analyticon** - Amazon-wide Conference on Data Engineering, Science and Analytics
Challenge: "Visualizing a Sustainable Tomorrow"
Approach: Multi-dataset integration with real NASA GISTEMP, NOAA, IRENA, and IEA data
Goal: Inspire climate action through compelling data storytelling
Features: Interactive visualizations,

correlation analysis, predictive modeling,
and accessibility-first design

Built with 💚 by @ygal and his best friend Amazon Q 🤖

Powered by AWS | NASA, IRENA, IEA, NSIDC & NOAA data